10. Determine basic feasible solution to the following transportation problem using North west Corner rule.

		Sinks					Supply
		A	*B*	*C*	*D*	*E*	
Origins	*P*	2	11	10	3	7	4
	Q	1	4	7	2	1	8
	R	3	9	4	8	12	9
Demand		3	3	4	5	6	

11. Find the initial basic feasible solution of the following transportation problem:

	I	II	III	Demand
A	1	2	6	7
B	0	4	2	12
C	3	1	5	11
Supply	10	10	10	

Using (i) North West Corner rule

(ii) Least Cost method

(iii) Vogel's approximation method

12. Obtain an initial basic feasible solution to the following transportation problem by north west corner method.

	D	*E*	*F*	*C*	Available
A	11	13	17	14	250
B	16	18	14	10	300
C	21	24	13	10	400
Required	200	225	275	250	

10.2 Assignment Problems

Introduction:

The assignment problem is a particular case of transportation problem for which more efficient (less-time consuming) solution method has been devised by KUHN (1956) and FLOOD (1956). The justification of the steps leading to the solution is based on theorems proved by Hungarian Mathematicians KONEIG (1950) and EGERVARY (1953), hence the method is named Hungarian Method.

Suppose that we have '*m*' jobs to be performed on '*n*' machines . The cost of assigning each job to each machine is C_{ij} (i =1,2,…,n and j = 1,2,…,n).Our objective is to assign the different jobs to the different machines(one job per machine) to minimize the overall cost.This is known as **assignment problem.**

The assignment problem is a special case of transportation problem where the number of sources and destinations are equal .Supply at each source and demand at each destination must be one. It means that there is exactly one occupied cell in each row and each column of the transportation table . Jobs represent sources and machines represent destinations.

10.2.1 Definition and formulation

Consider the problem of assigning n jobs to n machines (one job to one machine). Let C_{ij} be the cost of assigning i^{th} job to the jth machine and x_{ij} represents the assignment of ith job to the jth machine.

Then, $x_{ij} = \begin{cases} 1, \text{ if } i^{th} \text{ job is assigned to } j^{th} \text{ machine} \\ 0, \text{ if } i^{th} \text{ job is not assigned to } j^{th} \text{ machine.} \end{cases}$

		Machines				Supply
		1	2	…	*n*	
Jobs	1	(x_{11}) C_{11}	(x_{12}) C_{12}	…	(x_{1n}) C_{1n}	1
	2	(x_{21}) C_{21}	(x_{22}) C_{22}	…	(x_{2n}) C_{2n}	1
	⋮		⋮	⋮	⋮	⋮
	m	(x_{ij}) C_{n1}	(x_{ij}) C_{n2}	…	(x_{ij}) C_{nn}	1
	demand	1	1	…	1	

x_{ij} is missing in any cell means that no assignment is made between the pair of job and machine.(*i.e*) $x_{ij} = 0$.

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